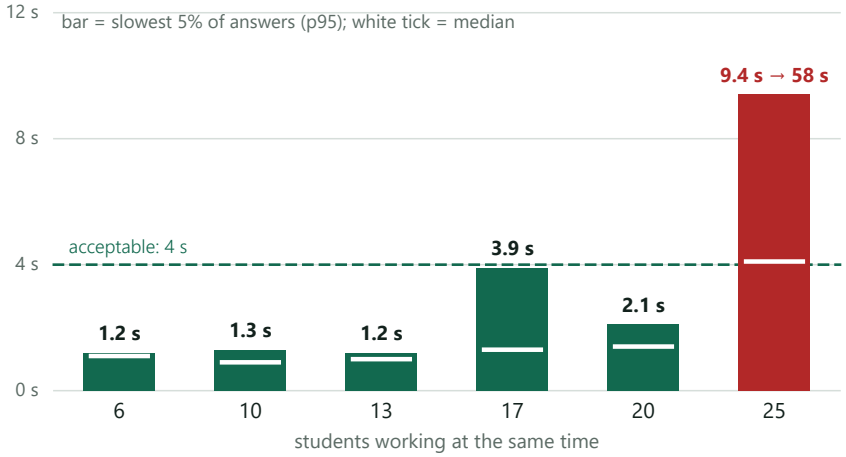
MCQ v3 load test: can 35 students sit one test at once?

swirl-mcq.streamlit.app (Streamlit Community Cloud + live Supabase) · 15 Oct 2026 · up to 60 simulated students on a 30-question paper

Not yet safe for 35. Safe today: about 20 students at once.

Up to ~20 students working at the same time, answers save in about 1.5 s with zero errors. At ~23–25 the server hits a cliff and **the whole class** stalls, not just the extra students.

CAPACITY AT A REALISTIC PACE (ONE ANSWER EVERY 20–40 S), DEPLOYED BUILD



Reading the chart

- **≤ 13 students:** about 1 s per answer, indistinguishable from one student alone.
- **17–20 students:** median 1.3–1.4 s; the slowest answers reach 2–4 s. Still usable.
- **25 students:** median 4 s, then within a minute 47 s. Clock frozen, pages won't load.
- The cliff is the server's CPU share on Community Cloud. Supabase was not the limit.

WHAT THE TEST FOUND, AND WHAT WAS FIXED

Before the fix: 35 students, fast pace

- App stopped saving answers for **7 minutes**; **17 of 35** were disconnected in one 10-second burst. Only **15 of 35** submitted.
- **17 answers (11 students) showed as selected but were never saved**; 9 students were marked 3–10 points low with no warning.
- Errors appeared with just **5 students**.
- Cause: every student shared one HTTP/2 connection to Supabase, and simultaneous clicks corrupted it (MCQ-BUG-032).

After the fix: commit e3f7fe4, deployed

- HTTP/1.1 connection pool: **no errors and no lost answers** up to ~20 students.
- Only the countdown refreshes every 5 s; the paper redraws when the student clicks. Shared quiz data is read once for the class.
- One student: answer **1.04 s → 0.61 s**; refresh **1.19 s → 0.20 s**.
- Time-up for **34 students at once: every paper was handed in correctly**, even while overloaded.

NEXT STEPS: CHANGES NEEDED FOR 35 STUDENTS

1. **Redraw one question per answer, not all 30.** Give each question its own fragment, so a click no longer re-renders the whole paper and its context block. This is the biggest saving per click (MCQ-BUG-033).
2. **Tick the clock every 15 s instead of 5 s.** It is now the only timer: 35 students make 7 refreshes a second today, and this cuts that to about 2.
3. **Skip the repeat read after each save.** The paper re-reads the attempt the student just wrote, so each answer costs one extra database round trip.
4. **Host on a server with 1–2 dedicated vCPUs.** The limit is Community Cloud's small, shared CPU slice, and it can't be upgraded.
5. **Re-test before the real class:** the realistic ramp to 60 students and the 'everyone submits within one minute' run, with a target of under 3 s per answer and zero errors at 40 students.

Method. Each simulated student is a real Streamlit session over the app's own websocket protocol (loadtest/swarm.py): sign in, start, answer 30 shuffled questions (with an ~8 KB balance-sheet context, like Quiz 4), then submit or let time run out. Latency = click to updated screen; browser drawing time excluded. Every stored answer and score was checked against what each student clicked. Runs at 3–15 s per answer were stress tests (3–5× a real class). **Not measured:** the 'everyone presses Submit within one minute' run on the fixed build (the laptop went into standby); the time-up run covers the harsher case. All ZZTEST test data was deleted afterwards. Full details: loadtest/REPORT.md.